KDDM1 PROJECT 2026 · PHASE I

Ice Hockey Talent Scouting

Phase I: data loading, preprocessing, cleaning and exploratory data understanding — without using the target variable.

7	5	10k	68
source files	file formats	players / rows	raw columns

Phase I pipeline — preprocessing, cleaning, EDA, no target use

1 Load

7 source files in 5 different formats (TSV, CSV, XLSX, PKL, JSON). Each loaded with format-specific parsing logic.

2 Merge

Outer join on international_id / medical_id. Mom's notes matched via (first_name, last_name) after deduplication.

3 Clean

Normalise units, fix typos, flag missing and invalid values. No blind imputation in Phase I.

4 Explore

EDA on the merged 10,000-player dataset: missingness, domain validity, correlations and visual patterns — without target use.

⚠ Noise / free-text / ID-like fields are documented as non-direct features. Unknown values are handled cautiously, not blindly filled.

Loading & Standardization

File	Format	Key Quirk
identity_card_0.tsv	TSV	Column names in Latin; IDs as Roman numerals
identity_card_1.csv	CSV	English columns; 9,112 players (complements id0)
performance.tsv	TSV	Typo: penality_minutes → penalty_minutes
medical_information.xlsx	XLSX	Junk metadata row + split header across rows 1–2
hockey_scouting_notes.csv	CSV	Clean; includes position, experience, scout notes
contracts_competition.pkl	PKL	Loaded from pickle; joined on international_id
moms_notes.json	JSON	No ID field; name-based matching required

Roman Numerals

CCLXXV → 275
custom ID parser

Latin → English

praenomen → first_name
cognomen → last_name

Excel Header

manual header reconstruction
unit parsing for medical data

Merging Strategy & Unknown Data

10,000	68	7	5
players in master dataset	columns after merging	source files loaded	file formats handled

Merge Pipeline

id0 + id1 → outer join on international_id + medical_id

- Coalesce duplicate columns (first_name, age, birth_city…)
- + performance.tsv on international_id
- + medical_information.xlsx on medical_id
- + hockey_scouting_notes.csv on international_id
- + contracts_competition.pkl on international_id
- + moms_notes.json via name + closest-age matching

Mom's Notes — Matching Strategy

- Problem: moms_notes.json has no stable ID column
- Unique names: direct match on (first_name, last_name)
- Ambiguous names: choose closest age, keep uncertainty visible
- Unmatched rows: kept as missing values, not deleted
- Missingness mechanism cannot be proven in Phase I (MCAR/MAR/NMAR only discussed cautiously)
- Processed output keeps invalid flags for later Phase II preprocessing

Numeric Correlations

Key Insights

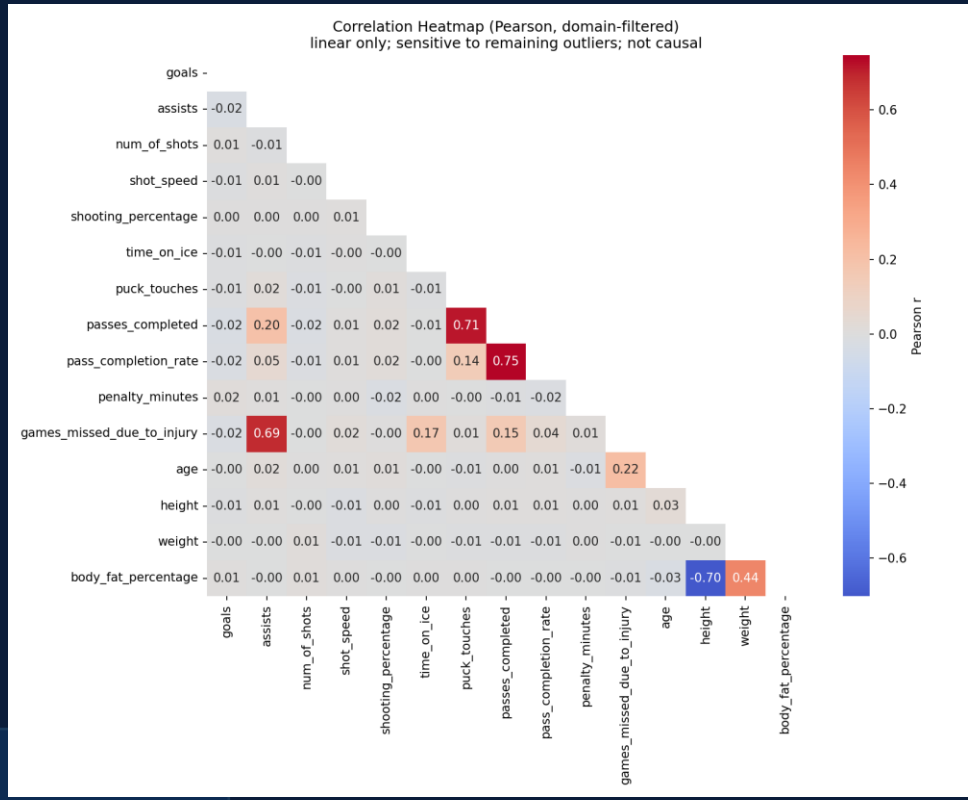
r = +0.75 passes_completed ↔ pass_completion_rate

r = +0.71 puck_touches ↔ passes_completed

r = −0.70 height ↔ body_fat_percentage

r = +0.69 assists ↔ games_missed_due_to_injury

Pearson is linear only; sensitive to outliers; no causal claims; high correlations warn about multicollinearity.



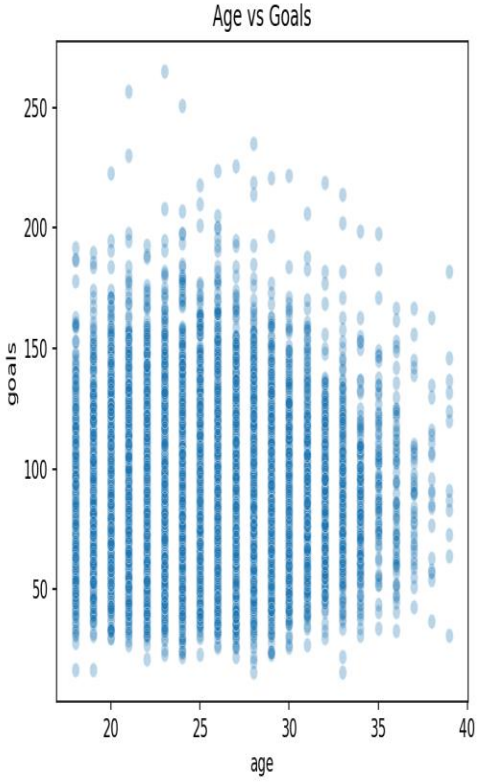
Scatter & Grouped Views

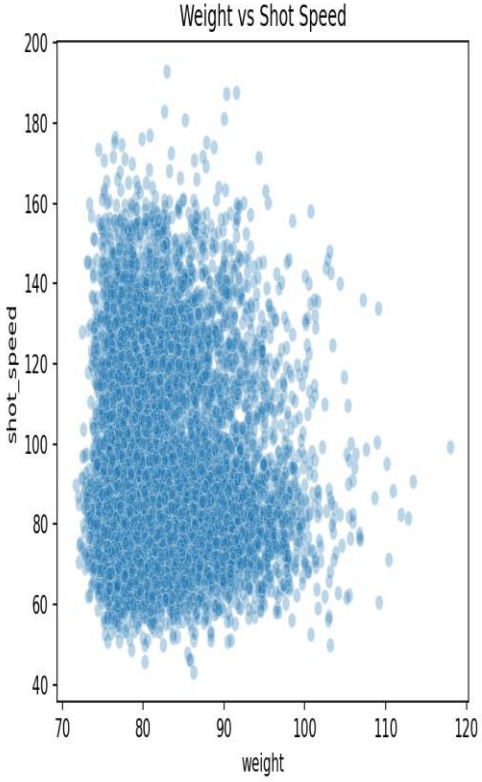
Visual Relationship Checks

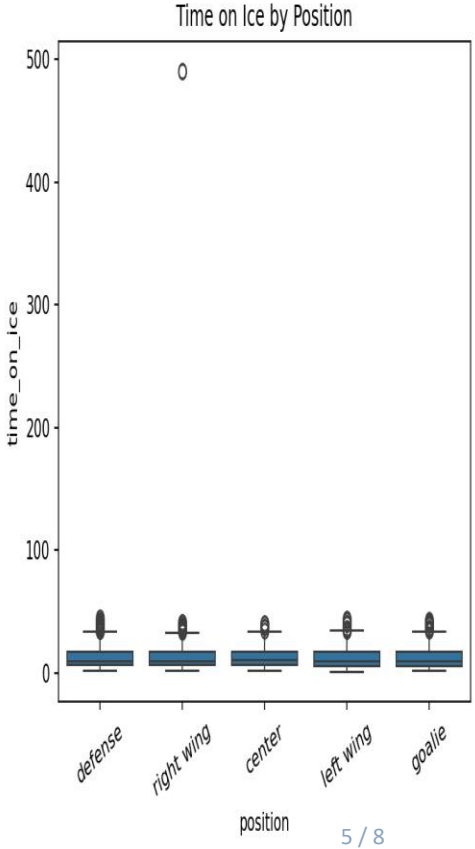
1. Age vs goals and weight vs shot speed show broad scatter rather than a clean deterministic rule.

2. Time on ice differs by position and contains one extreme value, so role-specific context matters.

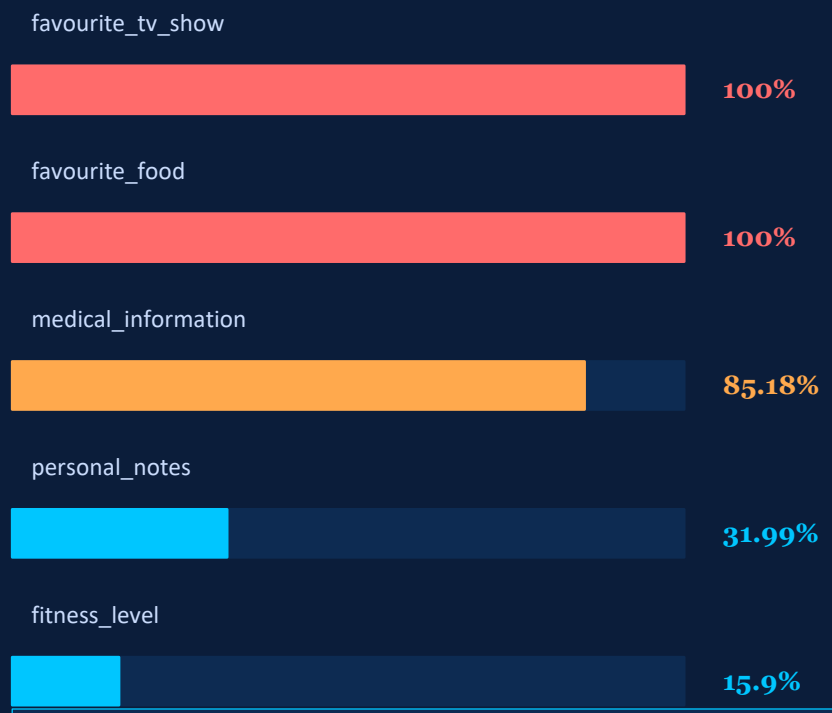
3. These plots support relationship exploration only — not causal conclusions.







Missing Values



Handling Strategy

- 100% missing: favourite_tv_show, favourite_food → unusable / drop candidates
- medical_information (85%): sparse free text; not a direct model feature
- fitness_level (15.9%): categorical → possible 'Unknown' label later
- Numeric missing: median imputation only after train/test split
- Date missingness: keep NaT + missing flag; no blind date imputation
- ~0.37% missing across 50+ performance columns: likely one source record absent, not random noise
- Mechanism check is only evidence: MCAR/MAR/NMAR not proven in Phase I, but first assumptions tested.

⚠ *No values are blindly imputed in Phase I. Missingness is quantified and documented; final imputation belongs after the Phase II train/test split.*

Outliers & Invalid Values

Treatment by Type

- IQR outliers: flagged as unusual, not removed automatically
- Domain-impossible values: replaced with NaN in processed dataset
- Invalid flags preserve where impossible values occurred
- Examples: age up to 455, height up to 3.09m, save % up to 106%, time on ice 490 min
- Rare but plausible players are kept; no rows are deleted
- No winsorising/capping of impossible values

age max 455

invalid → NaN + flag

61 height violations

up to 3.09m → NaN

save % up to 106%

10 impossible values → NaN

time on ice: 490 min

right wing outlier → NaN

0 rows deleted

only values flagged

Phase I result: the data is usable, but only after explicit cleaning rules

Heterogeneous Sources

7 source files in 5 formats merged into a 10,000-row, 68-column master dataset with custom parsing and careful joins.

Missingness is Structured

Two columns are 100% missing; medical_information is 85.18% missing. MCAR/MAR/NMAR is tested, but strategy stays cautious for detailed analysis in phase 2 if necessary.

Invalid ≠ Unusual

Domain-impossible values such as age 455 or height >3m are replaced with NaN and flagged; IQR outliers are not automatically removed.

Strong Associations

Top Pearson pairs: passes_completed/pass_completion_rate (r=+0.75), puck_touches/passes_completed (r=+0.71), height/body fat (r=−0.70).

Processed Output

phase1_processed_dataset.csv keeps all rows, replaces domain-impossible values with NaN, and adds 6 <column>_invalid_flag columns (68 → 74 cols total).

Phase II Next Steps

Impute after train/test split, handle text separately, reduce redundant features, and keep correlation caveats in mind.